

BOJ OIS Swap Rate Prediction

Model Evaluation & Hedge Ratio Report
11 instruments: M1-M8 (forward) + T12/T18/T24 (spot)

Data: 2007-03-05 to 2026-03-13 | OOS from: 2024-01-01

Part 1 — Model Evaluation

LightGBM | Walk-forward OOS | 3d & 5d horizons

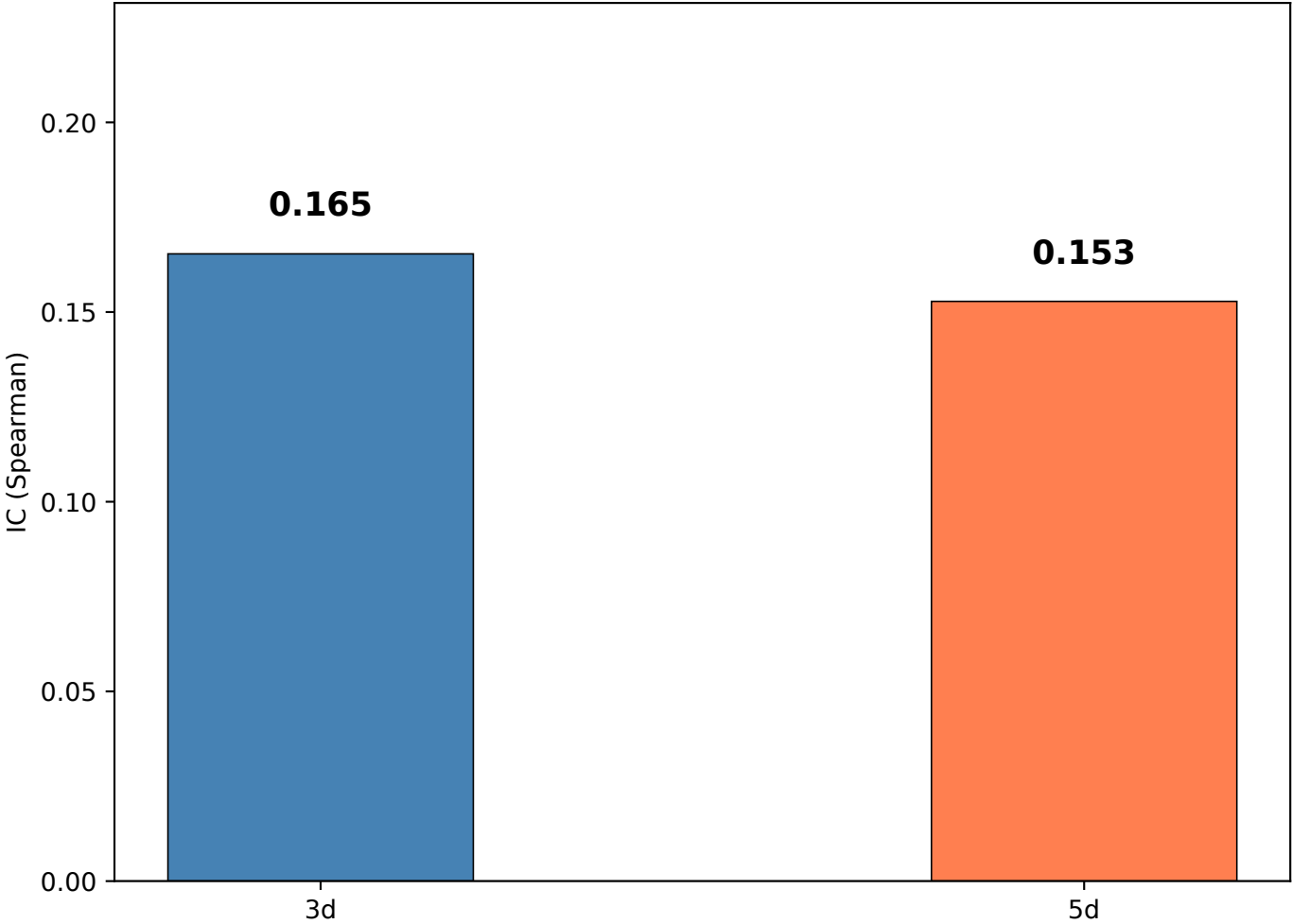
Data: 2007-03-05 to 2026-03-13 | OOS from: 2024-01-01

Overall OOS Performance (start: 2024-01-01)

Key Metrics

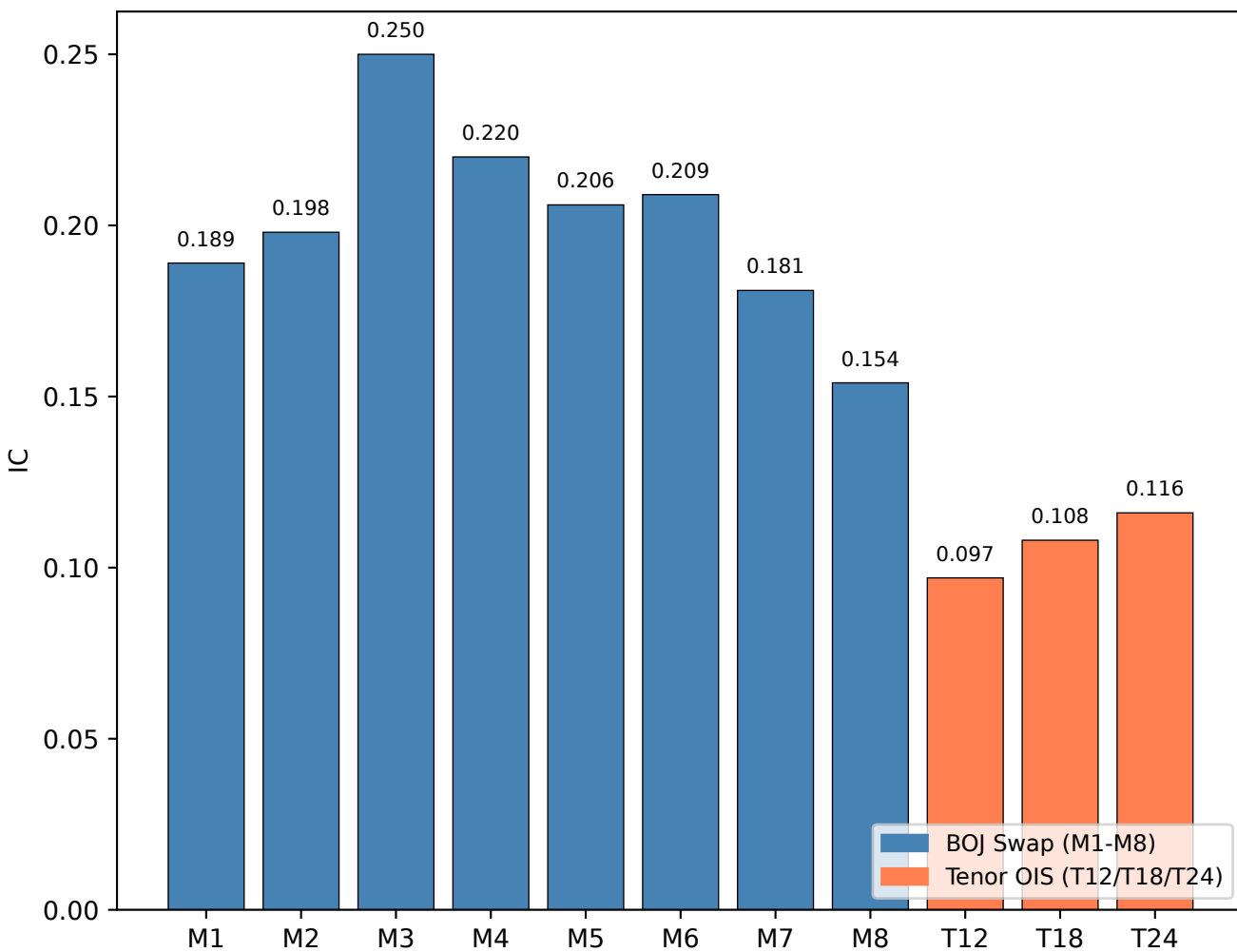
Metric	3d Horizon	5d Horizon
IC (Spearman)	0.165	0.153
RMSE (norm)	1.7063	1.9805
Direction Accuracy	52.3%	51.6%
Dir Acc (large move)	59.1%	61.8%
Instruments	11 (M1-M8, T12/T18/T24)	
Observations (OOS)	4708	4686
Walk-forward folds	9	9

OOS IC by Horizon

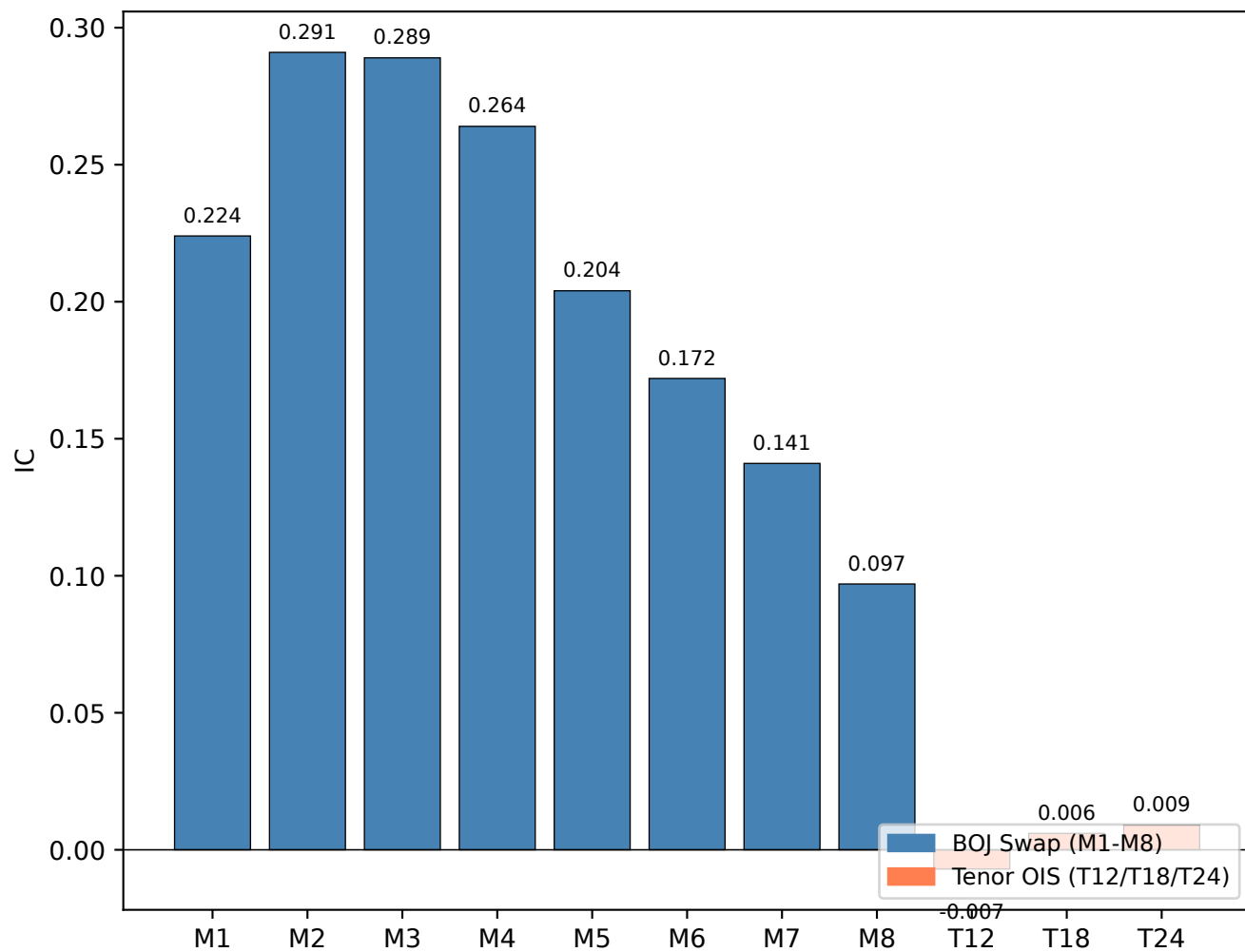


OOS IC by Instrument

3d Horizon

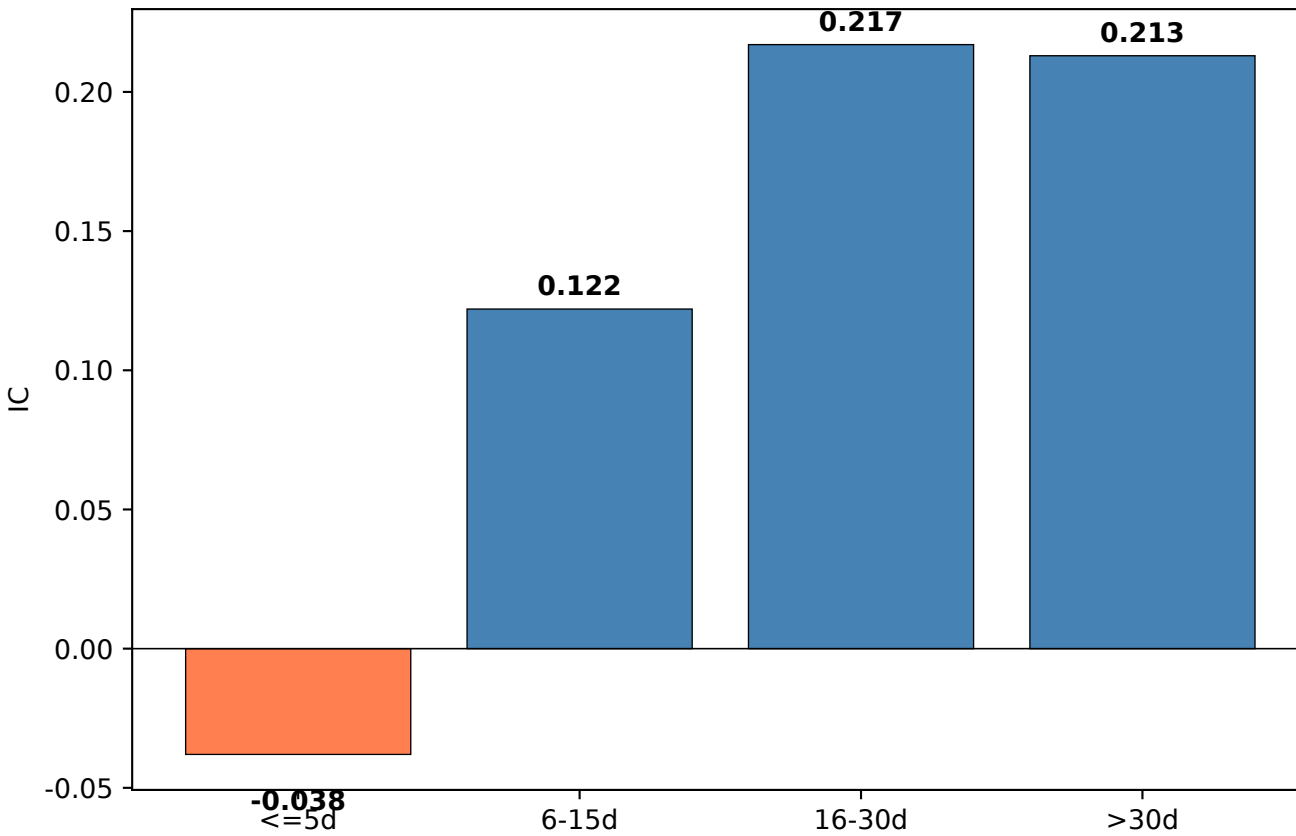


5d Horizon

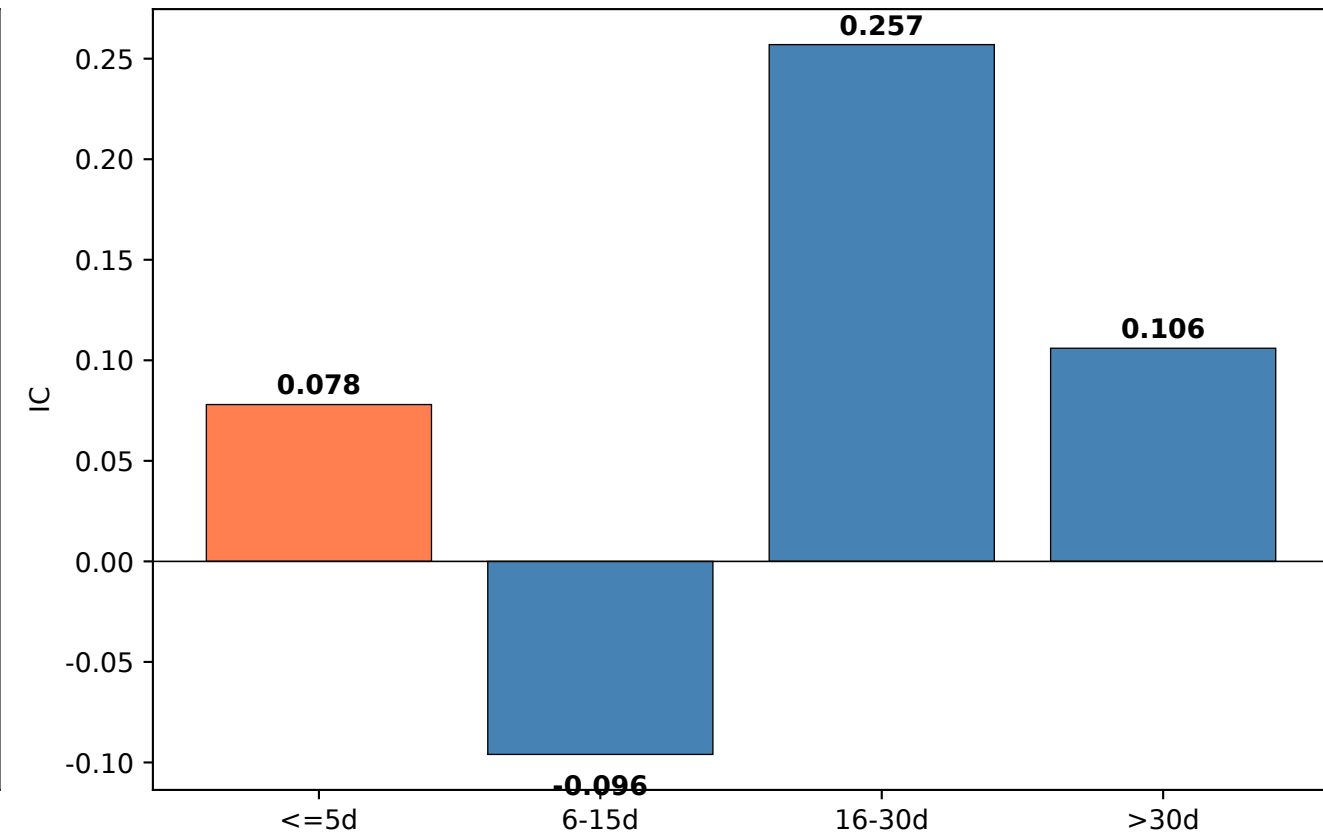


IC by Days-to-MPM & Walk-forward Fold Stability

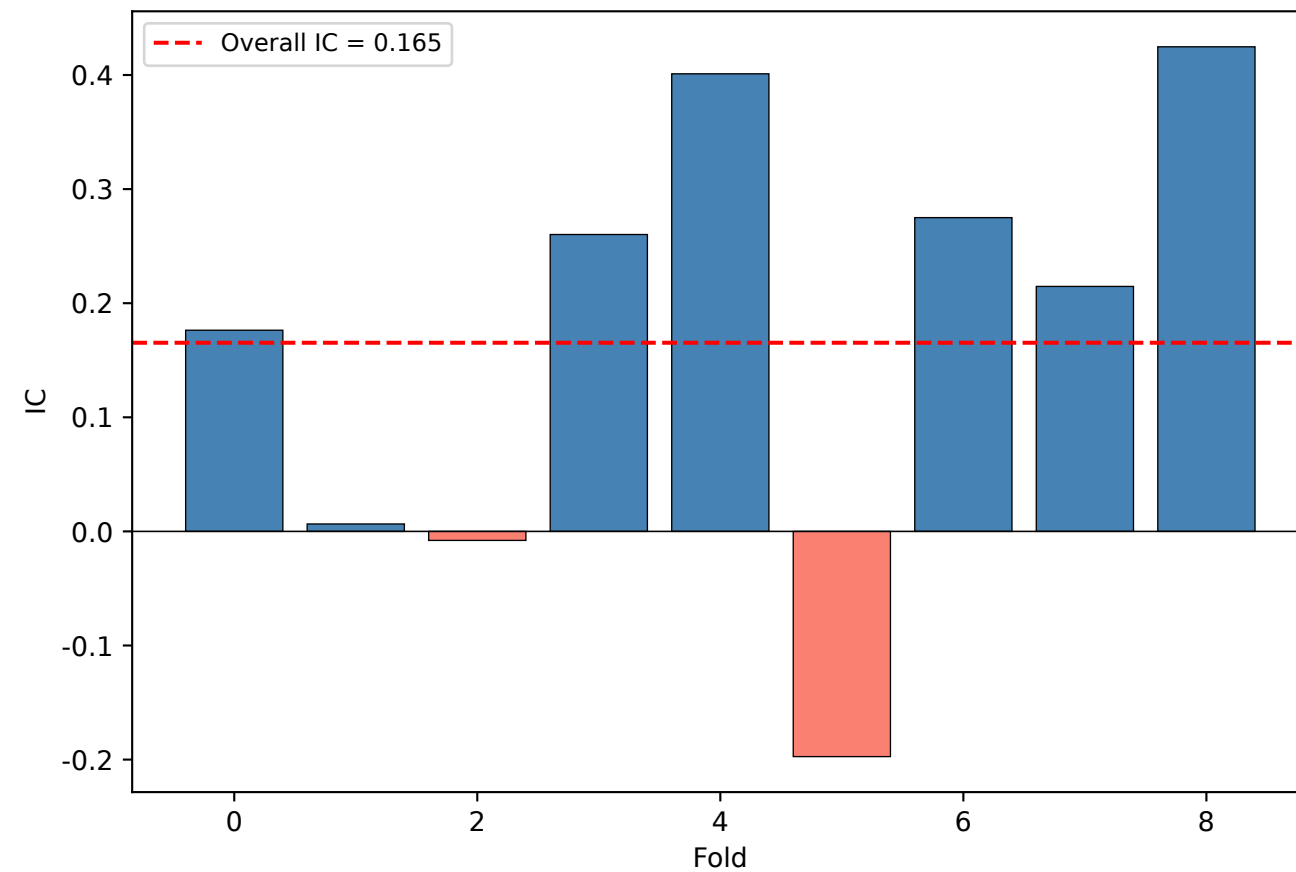
3d: IC by Days to MPM



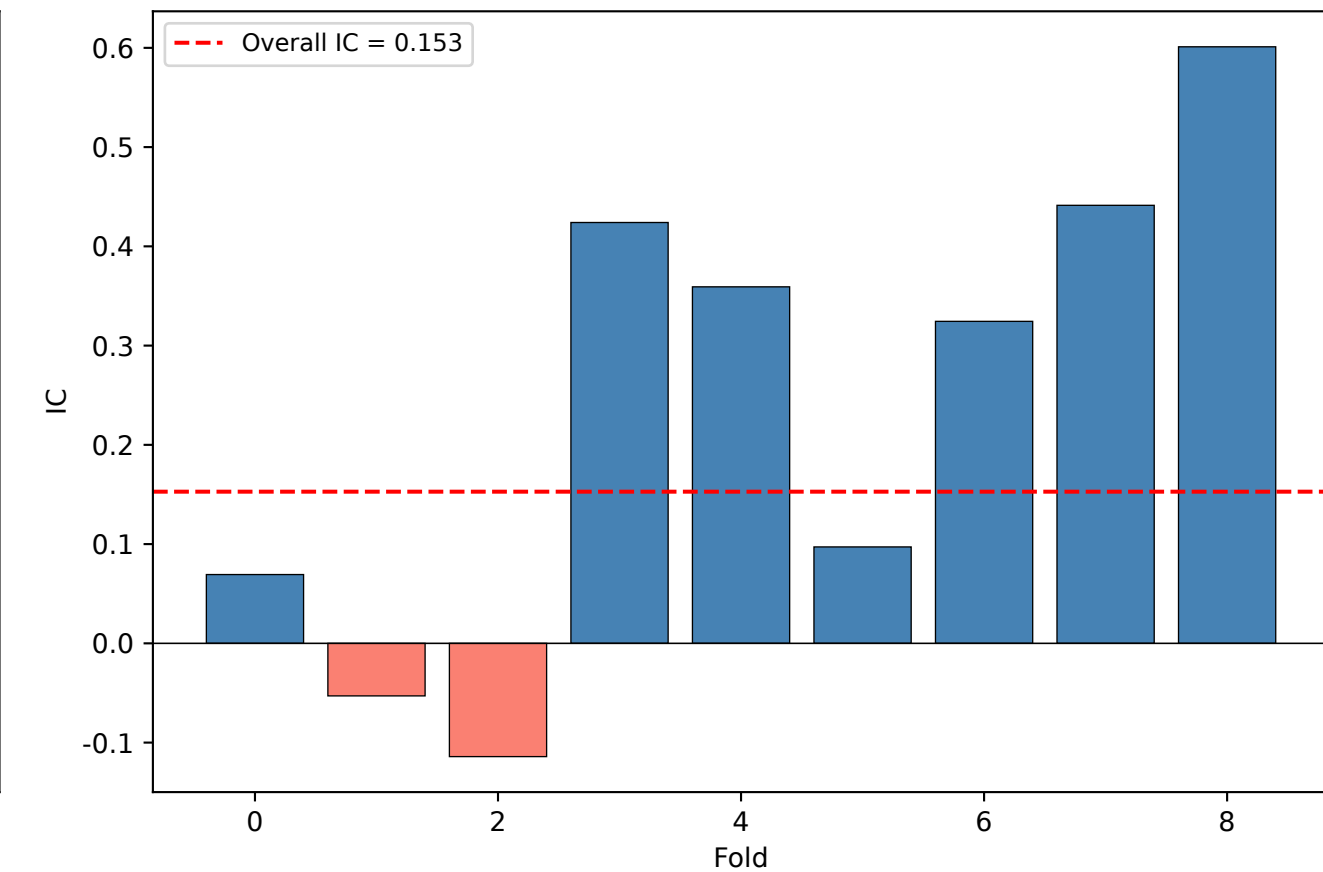
5d: IC by Days to MPM



3d: Fold IC

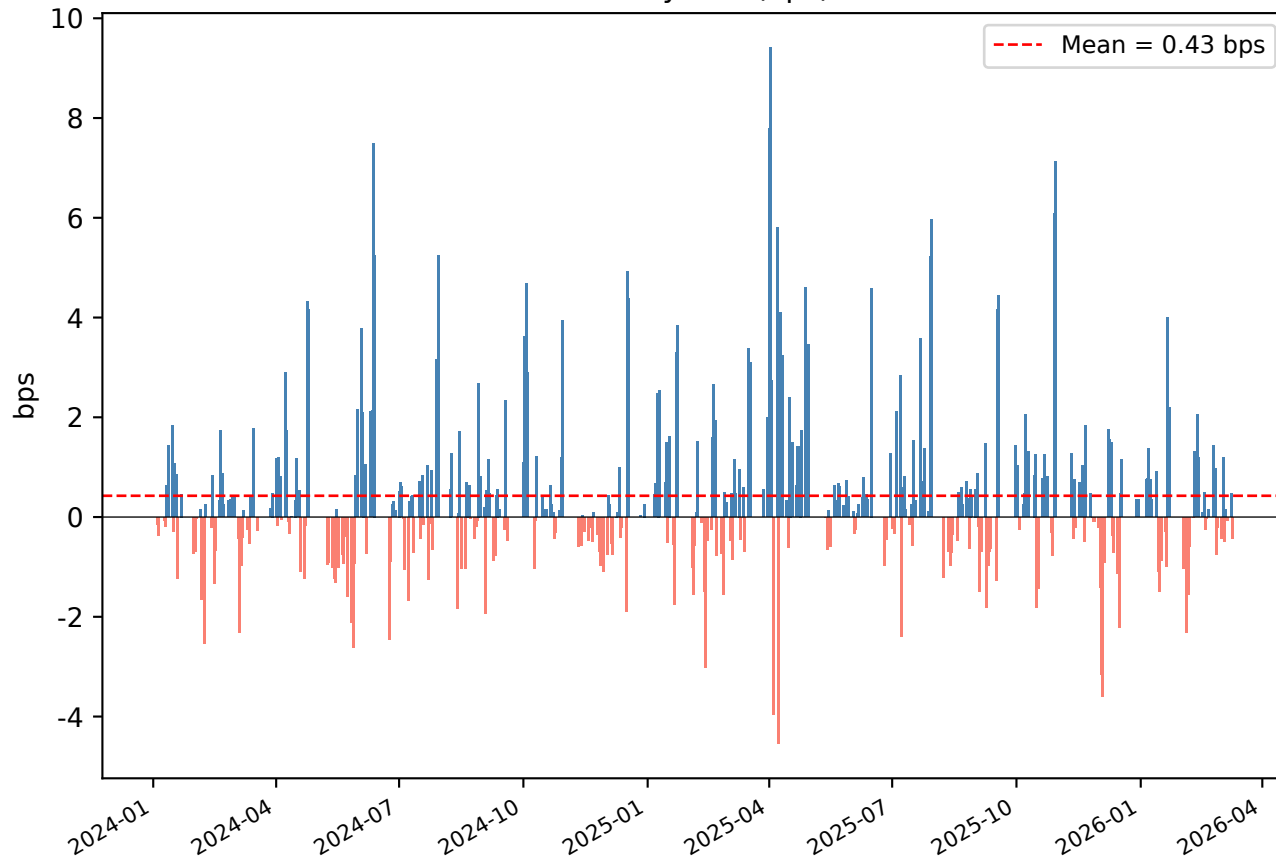


5d: Fold IC

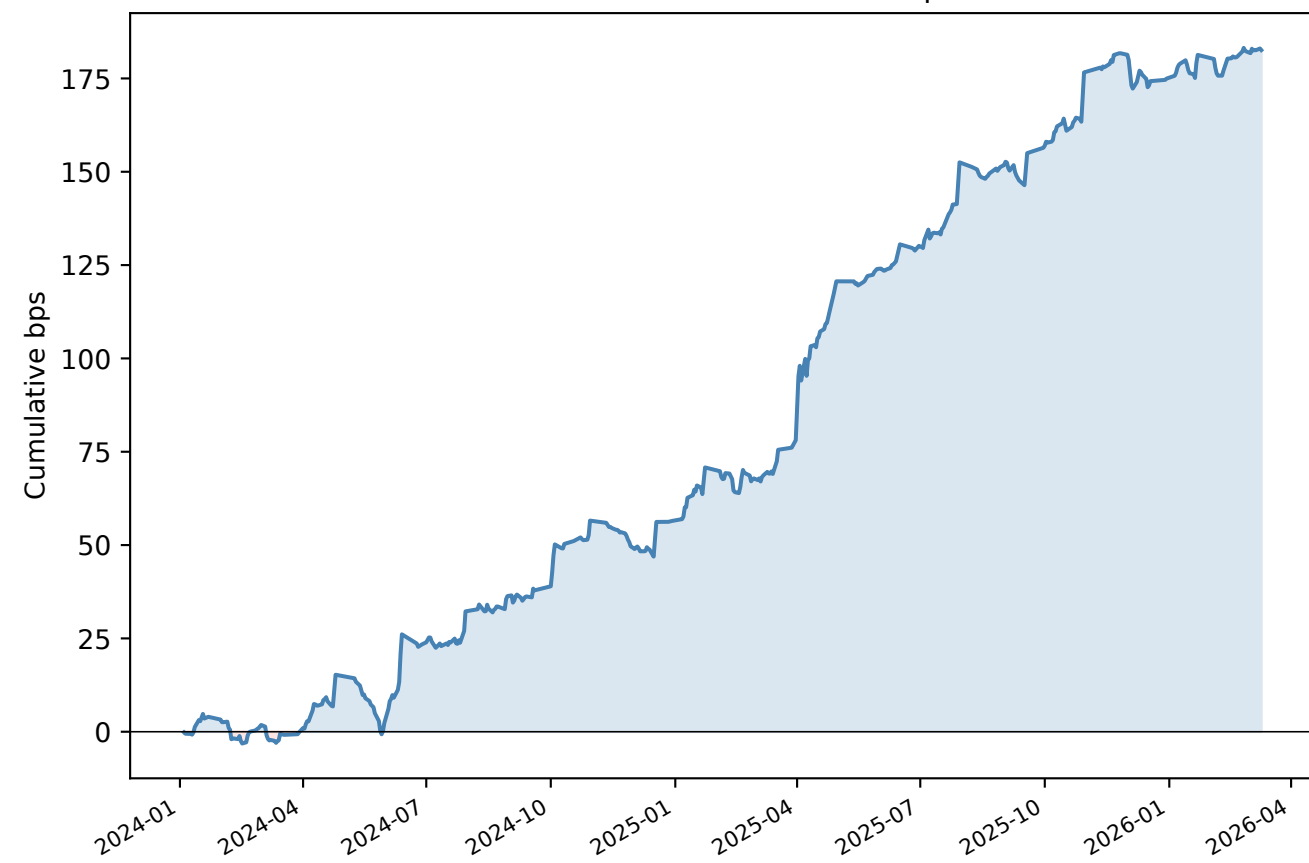


Cross-sectional P&L (Top4 Long / Bottom4 Short, daily)

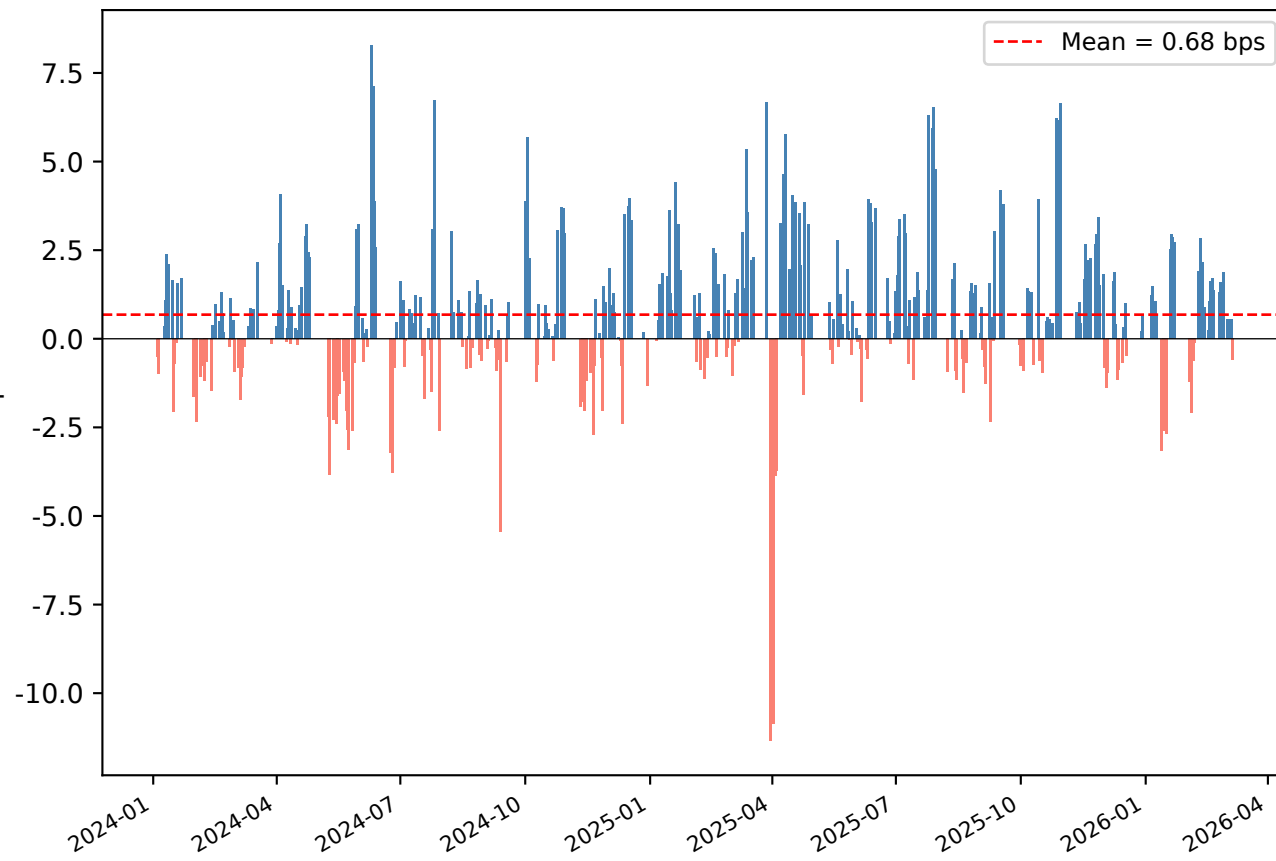
3d Daily P&L (bps)



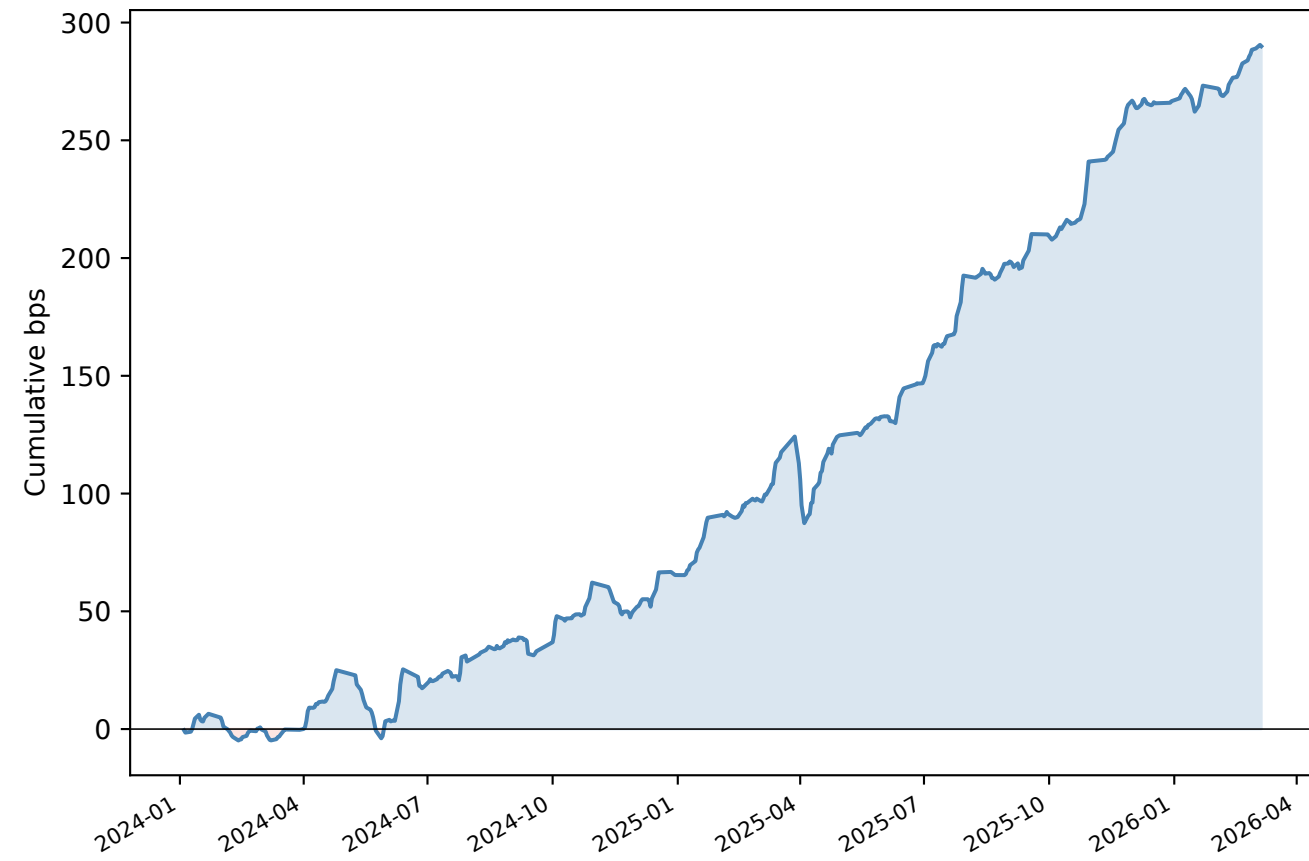
3d Cumulative P&L = 182.6 bps



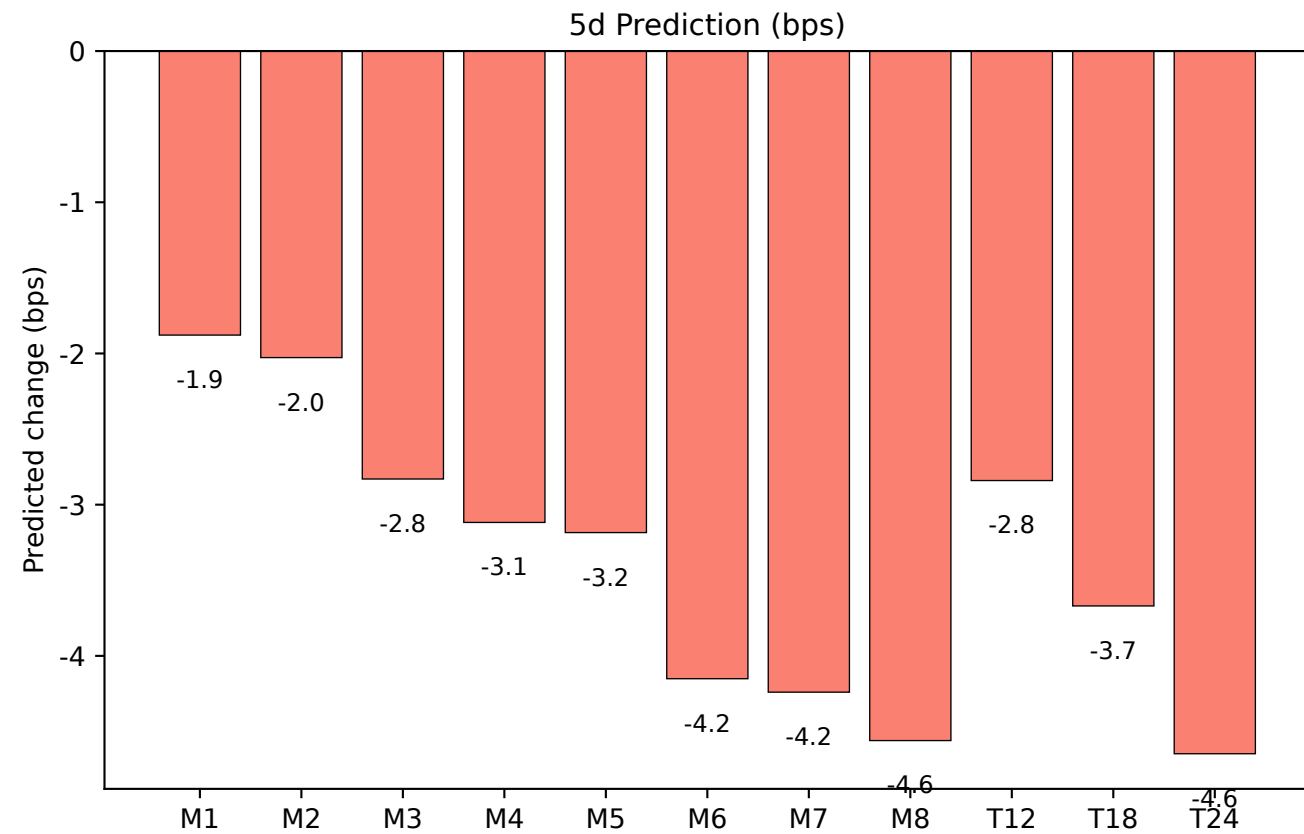
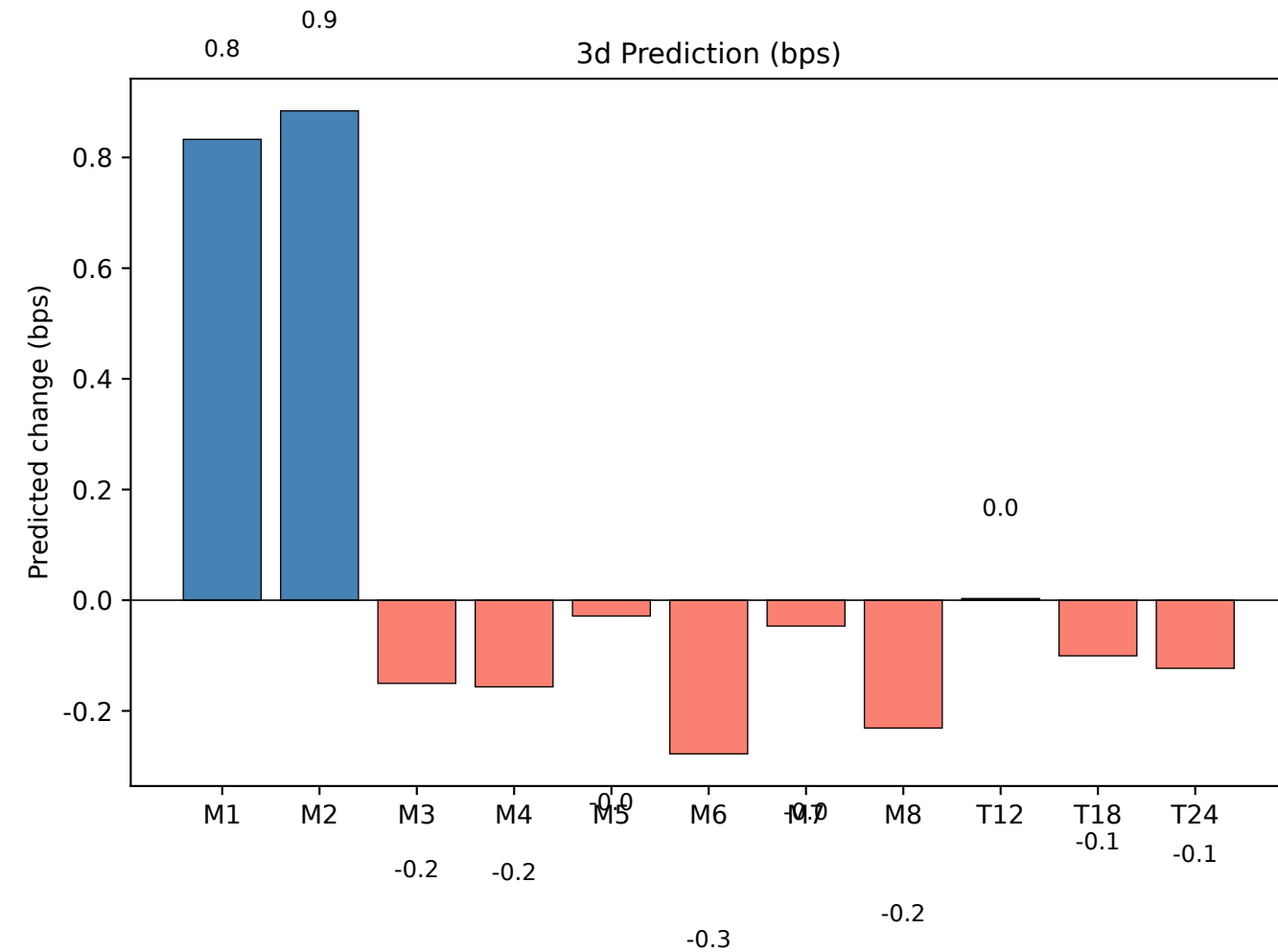
5d Daily P&L (bps)



5d Cumulative P&L = 289.9 bps



Latest Predictions (as of 2026-03-13)



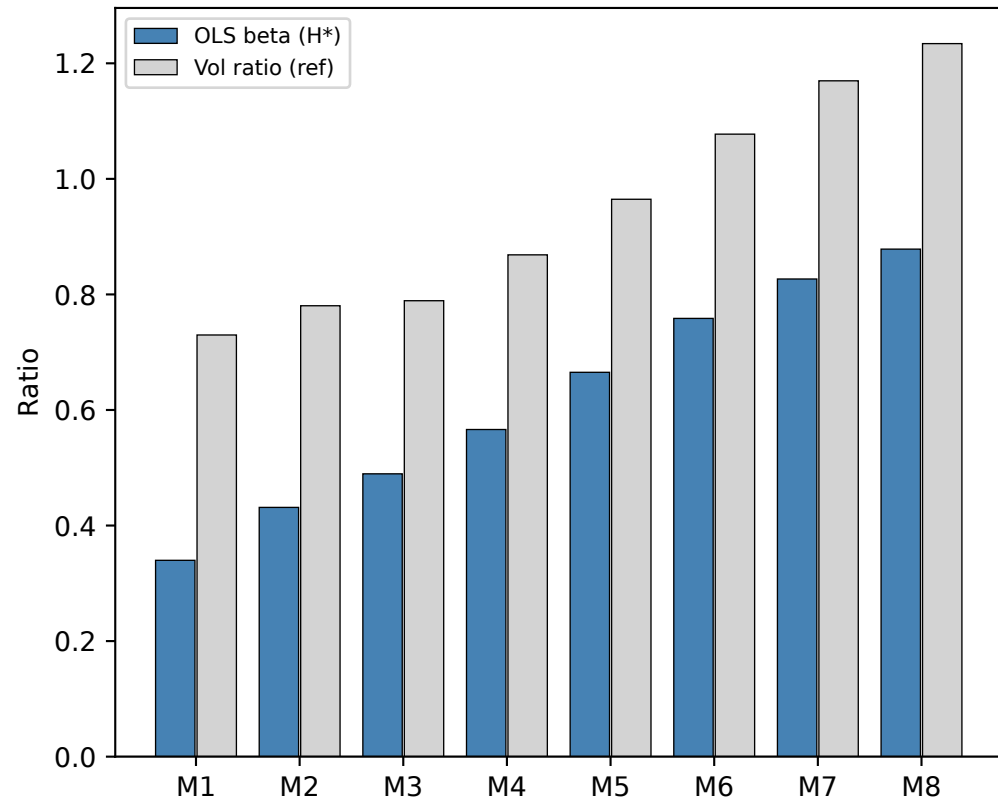
Part 2 — Hedge Ratio Analysis

OLS Minimum Variance Hedge | M1-M8 vs T12
avg meeting interval = 45.8d | avg M1 tenor = 2133.5d

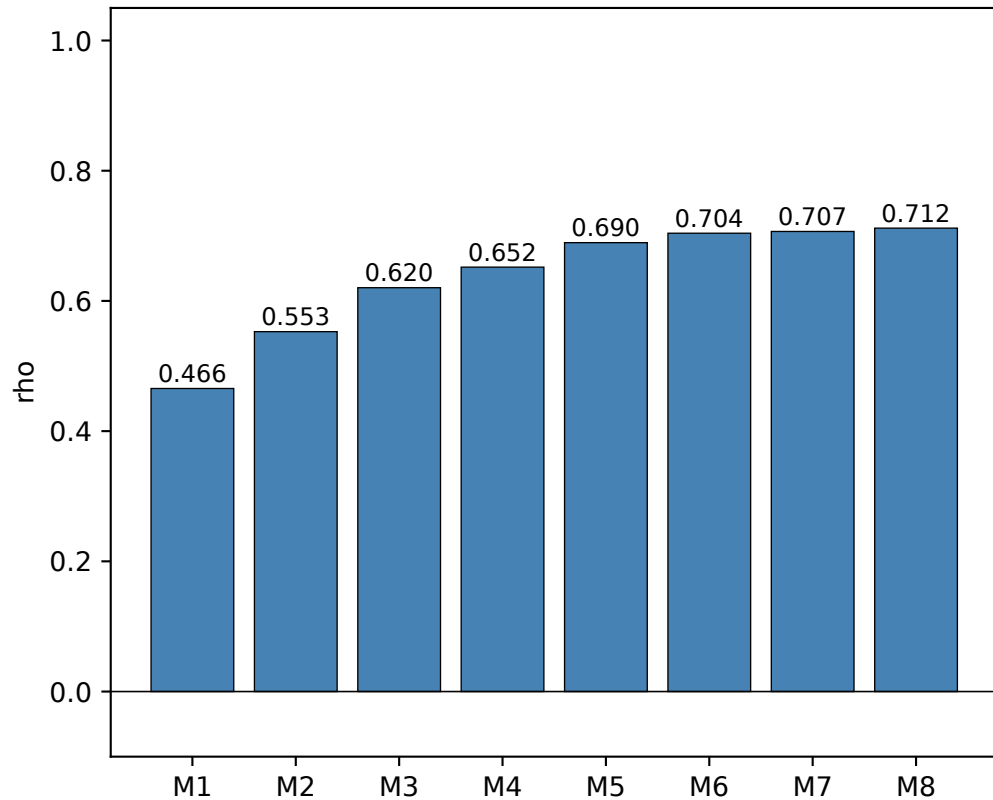
Data: 2007-03-05 to 2026-03-13 | OOS from: 2024-01-01

OLS Hedge Ratio vs T12 — Full Period

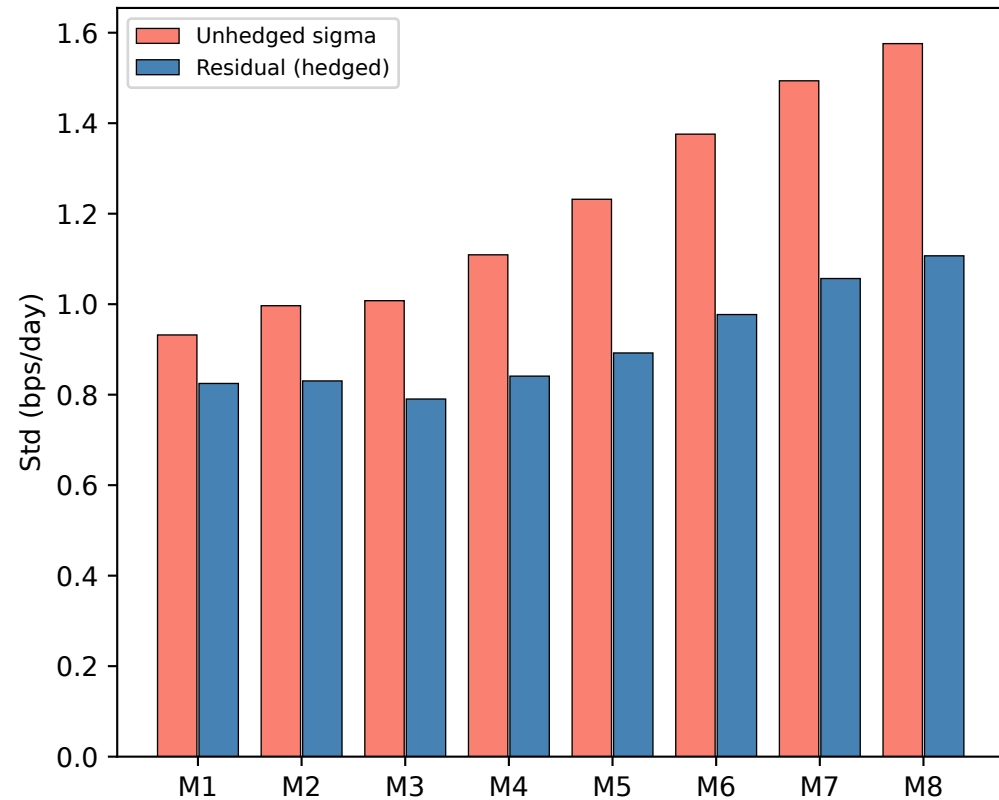
Hedge Ratio H^* vs Vol Ratio



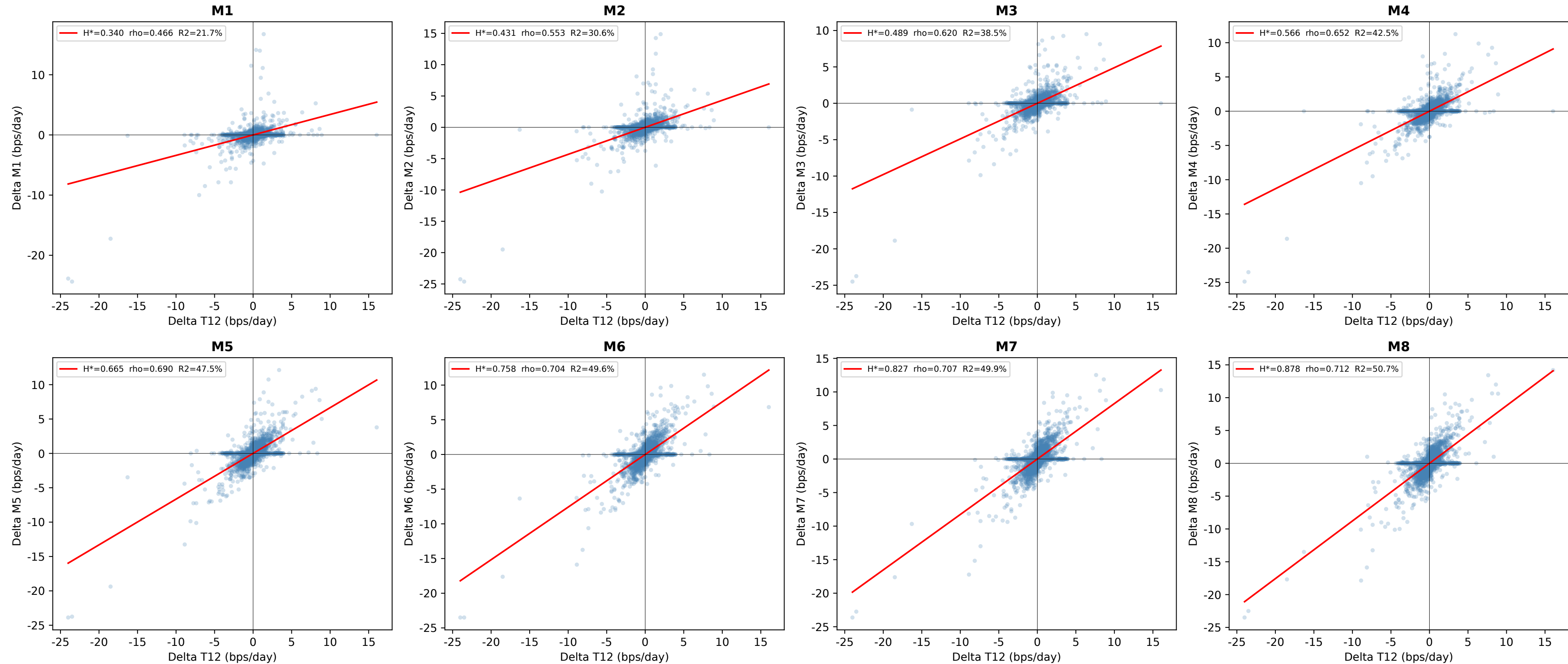
Correlation (ρ) with T12

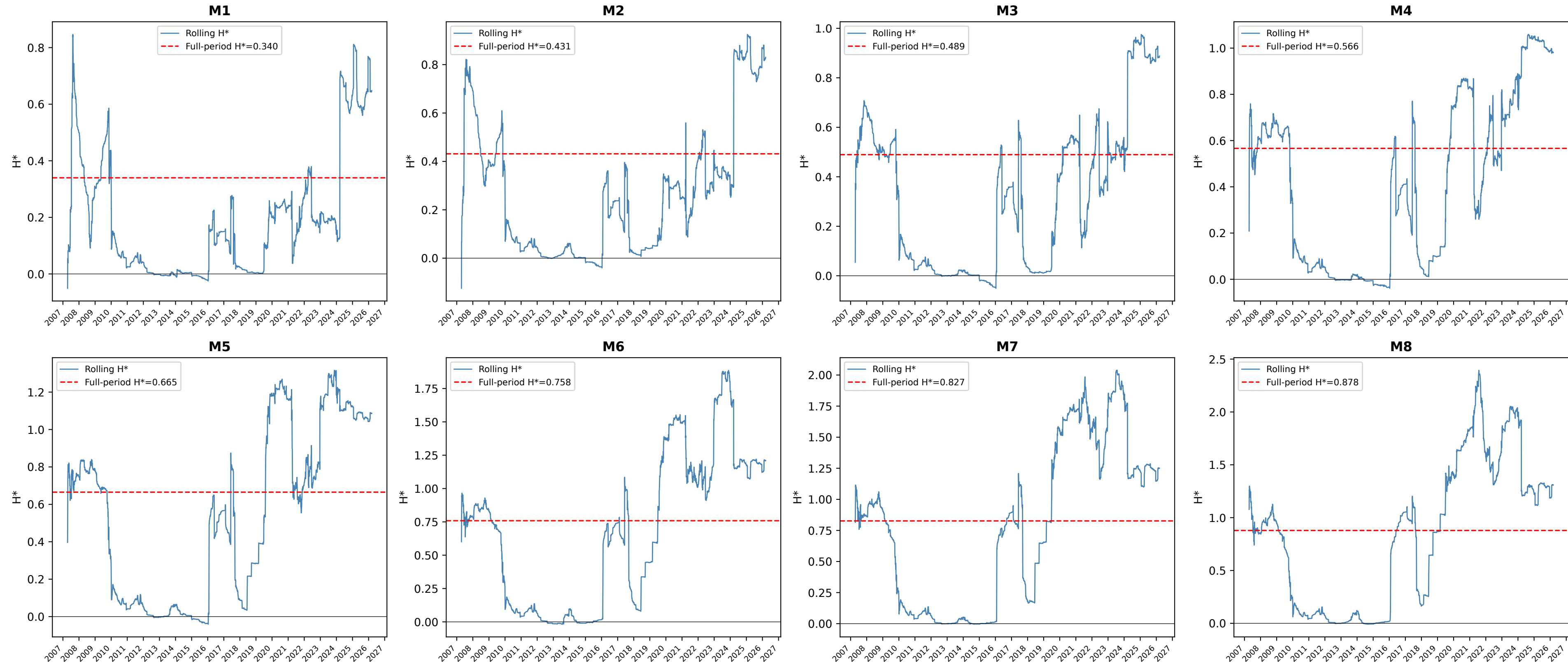


Risk Before vs After Hedge (bps/day)



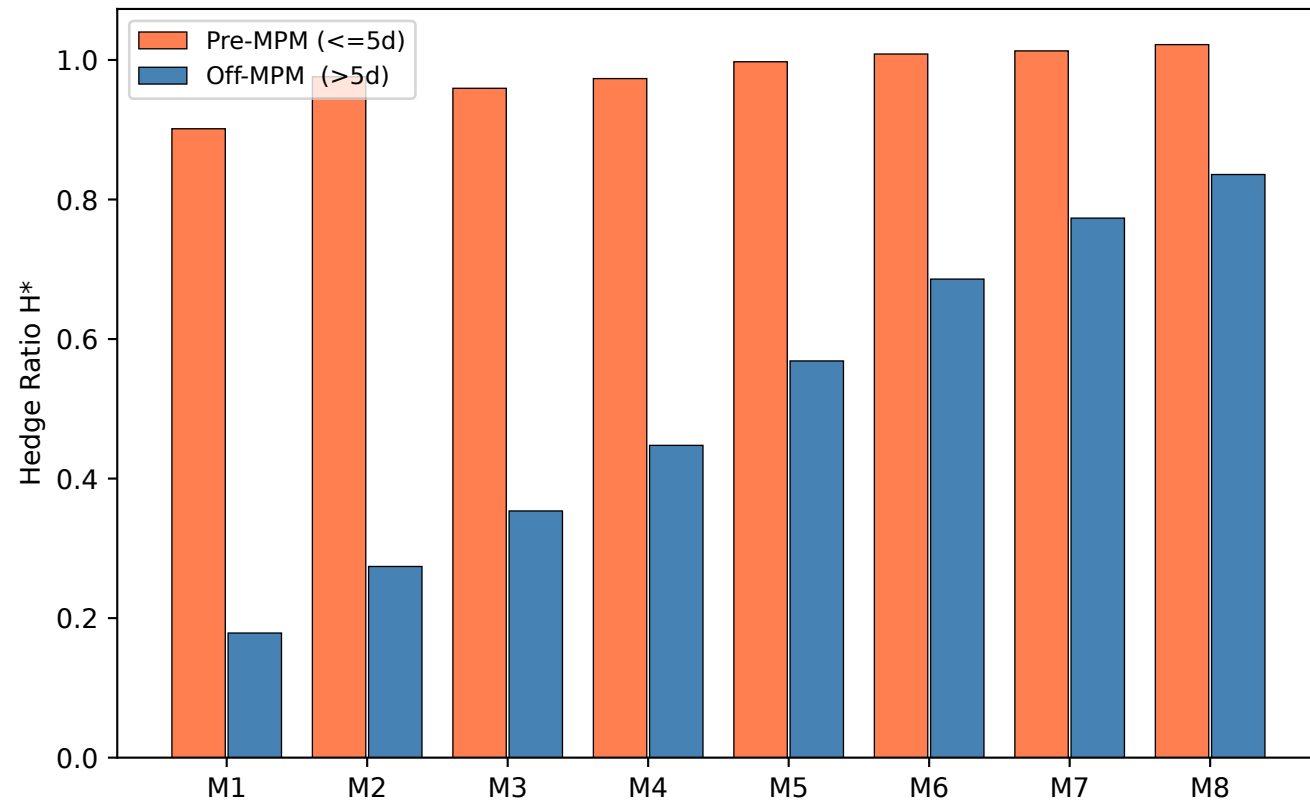
Daily Spread Change: M_n vs T12 (OLS regression line)



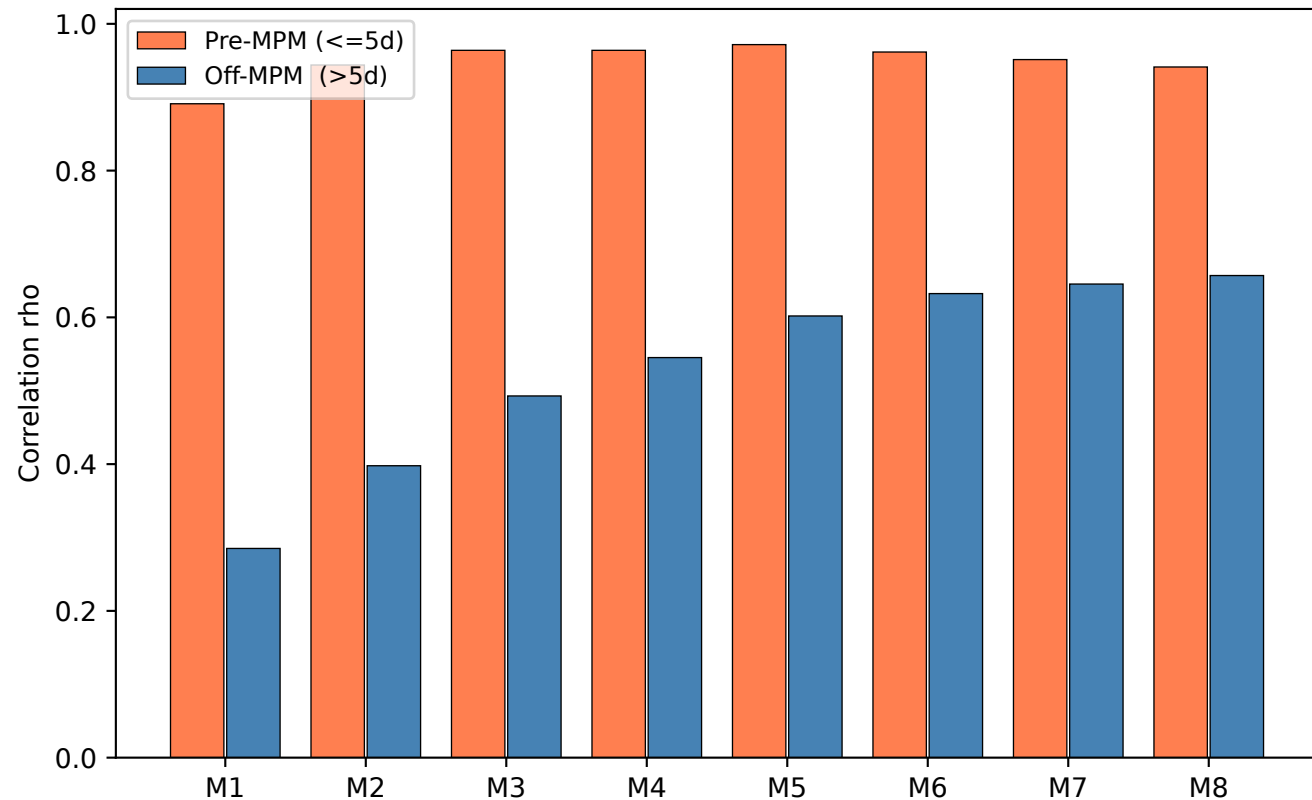
Rolling Hedge Ratio H^* (252-day window) — M_n vs T12

Hedge Ratio H^* and Correlation — Pre-MPM ($\leq 5d$) vs Off-MPM ($>5d$)

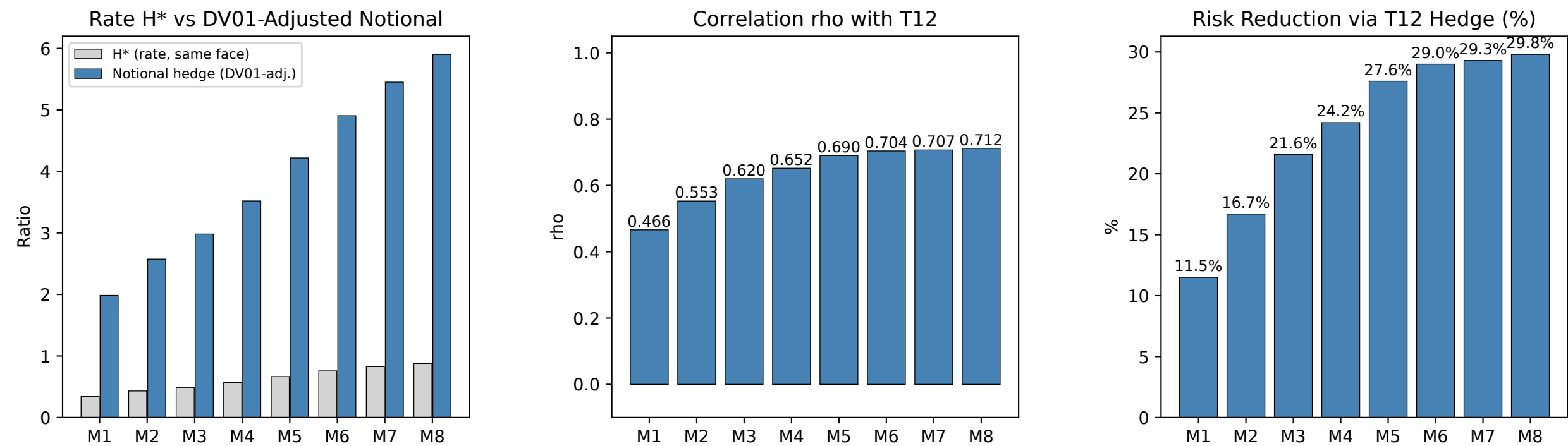
Hedge Ratio H^*



Correlation ρ



DV01-Adjusted Notional Hedge: M_n vs T12



DV01-Adjusted Notional Hedge Table
Notional hedge = H* x (tenor_Mn / 365d) Interpretation: for 1 unit of M_n, short [Notional hedge] units of T12

Instrument	Tenor (d)	DV01 ratio	H* (rate)	Notional hedge	rho	R2 (%)	sigma (bps)	Resid (bps)	Risk redn (%)
M1	2133	5.841	0.34	1.985	0.466	21.7	0.932	0.825	11.5
M2	2179	5.967	0.431	2.574	0.553	30.6	0.997	0.83	16.7
M3	2225	6.092	0.489	2.982	0.62	38.5	1.008	0.79	21.6
M4	2270	6.218	0.566	3.52	0.652	42.5	1.109	0.841	24.2
M5	2316	6.343	0.665	4.219	0.69	47.5	1.232	0.892	27.6
M6	2362	6.468	0.758	4.906	0.704	49.6	1.376	0.977	29.0
M7	2408	6.594	0.827	5.451	0.707	49.9	1.494	1.057	29.3
M8	2454	6.719	0.878	5.902	0.712	50.7	1.576	1.107	29.8